Deep Learning-Based Acoustic Analysis of Assamese Traditional songs with specific reference to Brajavali language present in Borgeet

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Abstract. This research presents a comparative study on the classification of a few Assamese traditional songs, Bihugeet, Kamrupiya Lokogeet, and the devotional song Borgeet, by employing Deep Learning Techniques, including DNN (Deep Neural Network), CNN (Convolutional Neural Network), and LSTM (Long Short-Term Memory). Among them, Borgeet is particularly significant as it is written in Brajavali language, a literary dialect that combines Assamese, Maithili, and Sanskrit languages. This makes Borgeet a valuable resource for multilingual study but at the same time development of an Automatic Speech Recognizer for Borgeet becomes challenging due to a lack of resources on digital platforms. In this article, a few preliminary experiments are carried out to measure different acoustic parameters of brajavali language along with the evaluation of MFCC values to classify all these geet (songs) using convolutional neural network. The model achieved the highest accuracy 91.95% and 94.95% F1 score in Borgeet.

Keywords: Automated Music Classification · DNNs · CNNs · LSTM models · MFCC feature extraction · Classical Music of India

1 Introduction

Assamese traditional music encompasses diverse genres, including Borgeet, Bihugeet, and Kamrupia Lokogeet, each representing unique cultural and thematic elements. Particularly, Borgeet maintains a significant importance as it was written in the fifteenth and sixteenth centuries by Mahapurush Srimanta Sankardeva and Madhavdeva, using Brajavali, a form of speech that incorporates elements of Maithili, Assamese, and Sanskrit [1]. Bihugeet emphasizes themes of love, nature, and celebration. Kamrupia Lokogeet reflects the vibrancy of Assamese daily life through its dynamic rhythms. The rich melodic variations and historical influences across these genres underscore the depth and diversity of Assam's musical heritage [2].

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The automated classification of traditional songs is crucial for preserving cultural heritage and serves as a linguistic and acoustic foundation for speech technology. Using spectral characteristics like MFCCs & algorithms like CNN, DNN & LSTM-RNN, new developments in deep learning have made it feasible to recognize audio effectively. Although previous studies have explored Indian music classification, there is a scarcity in research relating traditional musical data with automated speech recognition advancements in low-resource languages such as Brajavali. This research aims to close that gap. The goal is to develop accurate and reliable models capable of distinguishing between song genres, thereby facilitating the preservation, organization, and analysis of Assam's musical heritage.

2 Related Work

Research in audio and music classification has advanced considerably in recent years. Sheikh Fathollahi and Razzazi [3] applied Euclidean and cosine similarity measures along with CNN on datasets such as MABD, Emotify, and GTZAN, achieving classification accuracies of 83.79% and 86.5%. Jothimani and Premalatha [4] integrated CNN and LSTM networks to predict speech emotions, reporting accuracies ranging from 84.9% to 99.6% across four datasets.

Chakrabarty et al. [5] proposed a digital music library for Indian classical music, using note structures and fundamental frequencies to assess song similarity, with applications in music therapy. Hasib et al. [6] developed BMNet-5 for multiclass classification of Bangla music genres, achieving 90.32% accuracy and leveraging SHAP for interpretable results.

In order to recognize deepfake audio, Hamzaet al. [7] used MFCC-based parameters in both deep and machine learning (ML) algorithms, showing that support vector machines (SVM) performed better. Bubashait and Hewahi [8] compared CNN, LSTM, and DNN for urban sound classification, with LSTM achieving the highest accuracy of 90.15%. In 2023, the Multiscale Audio Spectrogram Transformer (MAST), was presented by Wentao Zhu and Mohamed Omar, performed better than the acoustic Audio Spectrogram Transformer (AST) in terms of effectiveness & feature representations. In 2022, Felicia Andayani et al. presented a research paper [9] to identify persistent patterns in audio signals and categorize emotions, with a hybrid LSTM & Transformer Encoder. They use MFCC to extract speech features. Due to the linguistic variety plus the scarcity of labelled or annotated voice data, developing a low-resource language ASR is a challenging task. In 2025 [10], A. Rishi Bharadwaj et al. developed a self-supervised multilingual model for low-resource languages based on Wav2Vec2-XLS-R-300, which serves as the foundation for the Telugu language ASR system. These research works demonstrate the strong performance of deep learning methods in audio and music classification, encouraging the creation of automated systems specifically for categorizing traditional Assamese music. Table 1 presents a consolidated review of prior research on audio and music classification.

Table 1. Critical review of related study

Author & Year	Domain/Dataset	Model Used	Features	Accuracy
Sheikh Fathollahi & Razzazi (2021)	MABD, GTZAN	CNN	MFCC	83.79–86.5%
Bubashait & Hewahi (2021)	Urban Sound Dataset	CNN, LSTM, DNN	Spectrograms	LSTM: 90.15%
Hasib et al. (2021)	Bangla Music Genres	BMNet-5 (Deep CNN) + SHAP	MFCC	90.32%
Jothimani & Premalatha (2022)	Four Speech Emotion Datasets	Hybrid CNN-LSTM	Mel Spectrograms	84.9–99.6%
Hamza et al. (2022)	Deepfake Audio Detection	SVM, CNN, DNN	MFCC	SVM: 98.83%
Proposed Study (2025)	Assamese Traditional Songs	CNN, DNN, LSTM-RNN	MFCC	**91.15%**

2.1 Contribution

In this research, the main feature parameter considered for audio categorization is Mel-frequency cepstral coefficients (MFCC). These three categories of songs, most importantly, the Borgeet, are written in Brajavali. In this work, we have performed acoustic analysis of Brajavali as a foundational step for Automatic Speech Recognition. So, from the perspective of "Assamese traditional music" the current study on Assamese music is considered to be cutting-edge. Through complete analysis and testing, our results conclusively demonstrate that the CNN model utilizing the MFCC feature values validates a substantially greater degree of accuracy when compared to its competitors, hence confirming its efficiency in the categorization of audio signals. Through this research authors have tried to determine the impact of different acoustic features on the classification performance of all three models.

3 Materials and Methods

In this article, Mel-Frequency Cepstral Coefficients are used as the primary parameters for classification. This model serves as a baseline for assessing the performance of more complex architectures. MFCCs are widely adopted in audio processing because they capture both temporal and spectral characteristics of sound by modeling the frequency distribution within a short-time segment. Feature extraction was performed using the Python library Librosa [11]. In addition, Mel-scaled filter banks were generated through librosa feature mel-spectrogram, providing an enriched representation of the audio signals. Table 2 below presents a summary of various attributes of Indian Classical Music (ICM). We are focusing on creating, constructing, and testing the best model configuration, ensuring performance checks for improved classification accuracy and Effectiveness. The flow diagram of our proposed model is shown in Figure 1.

We focused on creating, constructing, and testing the best model configuration, ensuring performance checks for improved classification accuracy and effectiveness, as illustrated in Figure 1.

Table 2. Core attributes of Indian Classical Music (ICM)

ICM Elements	**Explanation**
Raga	A melodic framework built from a set of notes, providing the framework for improvisation and composition.
Swaras (Notes)	In Indian classical music, a raga typically consists of 7 swaras (sa, re, ga, ma, pa, dha, ni), each note providing a unique harmonic identity and emotional expression.
Aaroh & Avroh	Aaroh indicates the progression of notes in an upward direction, while Avroh represents their descent.
Vadi & Samvadi	The Vadi is the most emphasized or central note in a raga, whereas the Samvadi acts as a complementary note, enriching the raga's melodic structure.
Gamakas	Gamakas refer to embellishments or oscillatory movements between notes that add depth and expressiveness.
Pakad	A Pakad is a distinctive sequence of notes that captures the essence of a raga and helps in its recognition.
Tala	A rhythmic cycle, defined by a specific number of beats, which provides temporal structure to the music.
Thaat	Thaat denotes the ten primary scale systems that serve as the basis for categorizing different ragas.

3.1 Description of the Dataset

A total of 190 audio recordings from three Assamese traditional genres- Borgeet (109), Bihugeet (37), and kamrupiya lokogeet (44) were gathered from public sources and archival repositories, featuring about 100 different speakers. Each recording is approximately 120 seconds long and digitized at a 22.05 KHz sampling rate. To enhance the dataset for training, each song was segmented into ten equal-duration segments, ensuring segments from the same audio file remained within the same subset to avoid data leakage. The dataset was split into 55% training, 25% validation, and 20% testing for CNN and LSTM-RNN models, while for DNN, the ratio was 80% training and 20% for combined testing/validation. Table 3 presents the number of audio files from our audio resource categorized by type, alongside their corresponding sample lengths. Figure 2 illustrates how the MFCC data is stored in a JSON file.

Table 3. Dataset Variants

Audio Class	**Number of Audio**	**Total Length (in sec.)**	**Sampling Rate (KHz)**
Assamese Borgeet	109	120	22.05
Bihugeet	37	120	22.05
Kamrupia Lokogeet	44	120	22.05

Figure 3 shows how the wave plot is load using LIBROSA. A Discrete Cosine Transform (DCT) [12] was utilized to successfully capture the inherent attributes of the audio signal, allowing subsequent analysis and classification on the basis

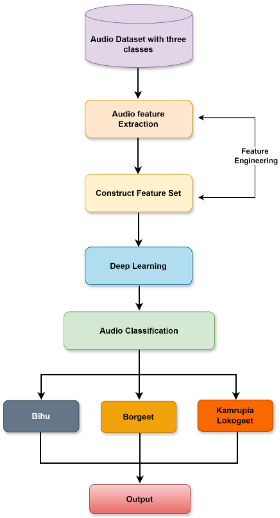

Fig. 1. Diagram demonstrating the suggested model's system

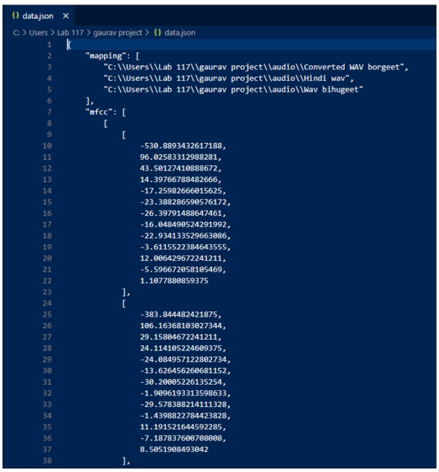

Fig. 2. Screenshot of MFCC feature values stored in JSON file

of the derived coefficients. This method successfully captures the inherent attributes of the audio signal, allowing subsequent analysis and classification on the basis of the derived coefficients.

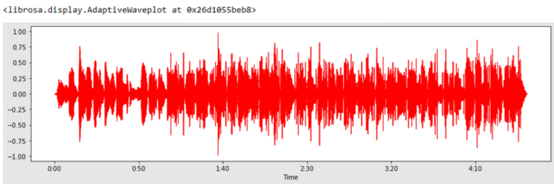

Fig. 3. Wave plot using LIBROSA

3.2 MFCC, or Mel-Frequency Cepstral Coefficients

The derivation of MFCC is grounded in a sequence of signal processing transformations that emphasize perceptually relevant aspects of human hearing. The audio signal is divided into short overlapping frames, typically 20–40 ms, enabling time-localized frequency analysis. The Short-Time Fourier Transform (STFT) is then applied to each frame to obtain the magnitude spectrum, expressed as

$$X(k, n) = |X_{\mathrm{STFT}}(k, n)|\,, \tag{1}$$

In equation (1), where $X_{\mathrm{STFT}}(k, n)$ represents the complex spectrum at frequency bin k for the n^{th} frame. Next, the magnitude spectrum is mapped onto the Mel scale with triangular filter banks that imitate the nonlinear frequency resolution of human hearing. The output of m^{th} Mel filter is given by equation (2).

$$H_m(n) = \sum_{k=0}^{N-1} X(k, n) \cdot W_m(k), \tag{2}$$

in the above expression $W_m(k)$ denotes the weighting function of the m^{th} Mel filter. This step effectively groups the spectral bins into perceptually meaningful frequency bands. To simulate the nonlinear loudness perception of the human ear, the Mel-filtered energies are logarithmically compressed as shown in equation (3).

$$M_m(n) = \log\big(H_m(n)\big). \tag{3}$$

Finally, a Discrete Cosine Transform (DCT) [9] is utilized on the log-filter bank energies to decorrelate the features and achieve a compact representation of the spectral envelope, which is shown in equation (4).

$$C_i = \sum_{m=0}^{M-1} M_m(n) \cdot \cos\left[\frac{\pi i}{M}(m + 0.5)\right], \tag{4}$$

In equation (4) C_i denotes the i^{th} cepstral coefficient and M is the total number of Mel filters. The DCT emphasizes the smooth spectral variations while suppressing noise-related fluctuations. The coefficients, known as Mel-Frequency Cepstral Coefficients (MFCCs), provide a compact and perceptually meaningful representation of the audio signal's spectral shape. In practice, only the first 12–13 coefficients are typically retained, as they capture the most relevant information for distinguishing between audio classes while discarding higher-order coefficients that primarily represent noise. MFCCs are widely utilized in speech, music, and environmental sound recognition beacause of their robustness and interpretability [13, 14].

3.3 Modeling

The custom datasets were employed to assess the performance of three different neural network architectures. The evaluation focused on multiple criteria, including model complexity and computational efficiency. Each architecture utilized the Adam optimizer with a learning rate of 0.0001, a batch size of 32, and an average of 100 epochs. Cross-entropy loss was minimized while employing early stopping based on validation performance to prevent overfitting.

DNN The study developed a deep neural network with 5 layers: input, 3 hidden intermediate layers, and an output layer. The output layer uses Softmax activation for multi-class classification, while the dropout layer reduces overfitting by randomly suppressing 30 percent of neurons. The model was trained over 100 epochs to establish a performance baseline. This baseline facilitates the subsequent evaluation and comparison of model performance across different configurations. Figure 4 illustrates the overall structure of the proposed Deep Neural Network.

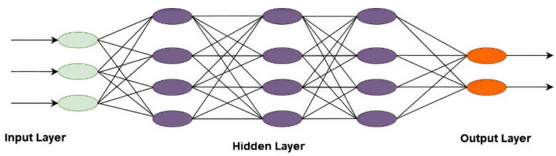

Fig. 4. Architecture of DNN model

CNN Convolutional neural networks (CNNs) incorporate essential architectural elements such as convolution and pooling layers, as well as input data format, filter size [15], and key hyperparameters, including learning rate, dropout rate, batch size, and training epochs, all vital for optimal performance.

Table 4. Summary of the proposed DNN architecture

Layer	**Output Dimensions**	**Number of Parameters**
Input Layer (Flatten)	(None, 13442)	0
Hidden Layer 1 (Dense)	(None, 512)	6,882,800
Dropout 1	(None, 512)	0
Hidden Layer 2 (Dense)	(None, 256)	131,300
Dropout 2	(None, 256)	0
Hidden Layer 3 (Dense)	(None, 64)	16,400
Dropout 3	(None, 64)	0
Output Layer (Dense, SoftMax)	(None, 3)	195

The CNN model uses a sequential design with three Conv2D layers, MaxPooling2D layers, and two dense layers, sliding learnable filters, performing element-wise multiplications, and aggregating feature maps. Figure 5 presents the comprehensive architecture of the CNN model. The architecture's layers and corresponding parameters are summarized in Table 5 [16].

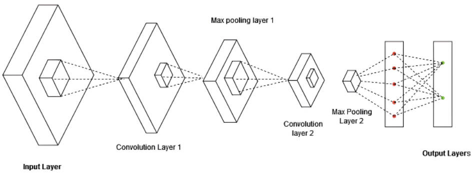

Fig. 5. Framework of the proposed CNN model

LSTM LSTM, or Long Short-Term Memory networks, are getting a lot of attention for audio classification because they can represent how data changes over time. The suggested architecture for the audio recognition task consists of four LSTM layers, a dense layer that is fully connected, and an output layer that produces class probabilities. Model parameters are optimized using the Adam optimizer, ensuring efficient convergence during training. Unlike traditional recurrent neural networks (RNNs), LSTM units mitigate the vanishing gradient issue, facilitating effective learning over extended sequences while ensuring computational efficiency [17]. Figure 6 provides an overview of the LSTM-based model, and Table 6 summarizes its layer configuration and associated parameters.

Table 5. Overview of the layers and parameters of the CNN architecture

Layer	Output Shape	Trainable Params
Conv1	(None, 515, 11, 64)	640
MaxPool1	(None, 258, 6, 64)	0
BatchNorm1	(None, 258, 6, 64)	256
Conv2	(None, 256, 4, 32)	18,460
MaxPool2	(None, 128, 2, 32)	0
BatchNorm2	(None, 128, 2, 32)	128
Conv3	(None, 127, 1, 32)	4,120
MaxPool3	(None, 64, 1, 32)	0
BatchNorm3	(None, 64, 1, 32)	128
Flatten	(None, 2048)	0
Dense1	(None, 64)	131,130
Dropout	(None, 64)	0
Output (SoftMax)	(None, 3)	195

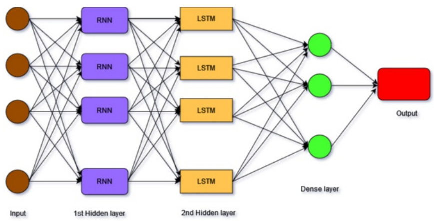

Fig. 6. Architecture of Proposed LSTM model

Table 6. Layer-wise summary of the proposed LSTM architecture

Layer	Output Dimensions	Trainable Parameters
LSTM Layer 1	(None, 517, 64)	19,960
LSTM Layer 2	(None, 64)	33,020
Dense Layer	(None, 64)	4,160
Dropout Layer	(None, 64)	0
Output Layer (Dense, SoftMax)	(None, 3)	195

4 Results and Discussion

The performance of each model in classifying audio samples was assessed using MFCC features extracted from our custom dataset. The baseline The overall accuracy of the baseline model was 89.04%, with F1-scores of 93.59%, 82.35%, and 71.18% for Borgeet, Bihugeet, and Kamrupiya Lokogeet, respectively. The overall model performance is shown in Figure 7 provides a detailed view of the F1 score of all three classifiers. Table 7 summarizes the comparative performance of the three models, DNN, CNN, and LSTM-RNN, across the three audio classes.

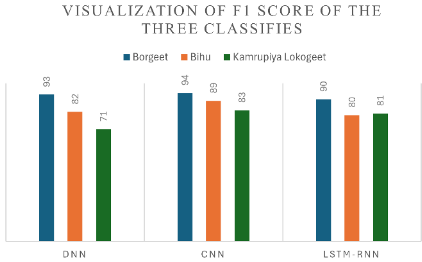

Fig. 7. Analysis of F1 performance in graphic form

The CNN model demonstrates the highest accuracy (91%) and competitive F1-scores across all classes, while the LSTM-RNN model provides efficient sequence modeling but slightly lower overall performance.

Table 7. Performance comparison of DNN, CNN, and LSTM-RNN models across audio classes

Model	Epochs	Accuracy (%)	Training Time (min)	Class	F1-score	Precision	Recall
DNN	100	89.04	7.05	Borgeet	0.94	0.90	0.97
				Bihugeet	0.82	0.87	0.77
				Kamrupiya Lokogeet	0.71	0.84	0.61
CNN	100	91.00	9.63	Borgeet	0.94	0.98	0.92
				Bihugeet	0.89	0.90	0.87
				Kamrupiya Lokogeet	0.83	0.76	0.92
LSTM-RNN	100	86.00	33.40	Borgeet	0.90	0.91	0.89
				Bihugeet	0.80	0.80	0.80
				Kamrupiya Lokogeet	0.81	0.80	0.83

5 Model Deployment

To enable practical usage of the proposed system, a graphical user interface (GUI) was developed to facilitate audio classification. The interface allows users to conveniently upload audio samples and obtain corresponding class predictions with minimal effort. This design emphasizes usability while maintaining accurate and efficient classification. The implemented interface's structure is shown in Figure 8

6 Limitations

The findings from the experiment are promising, but there are still some limitations: Dataset size: The dataset consists of 190 recordings, which is relatively

Fig. 8. Graphical user interface for audio classification.

small and may limit the generalization of trained models. Class imbalance: Learning can be affected by class imbalance due to uneven representation of song genres, with data balancing and augmentation strategies suggested as potential solutions. Feature scope: Only MFCCs were employed; the performance of the model might have been enhanced if we include spectral contrast, delta-MFCC, or chroma features. Generalization to speech: Future research will focus on spoken Brajavali and developing an end-to-end ASR pipeline with CNN-LSTM or Transformer architectures, in contrast to this study, which analyzed musical audio.

7 Conclusion

The suggested methodology continuously shows improvements in precision, accuracy, recall, and F1-score compared to previous methodologies & benchmark datasets. Among the three models evaluated, DNN, CNN, and LSTM, the CNN achieved the highest performance, outperforming DNN (89.04%) and LSTM (86.31%). The CNN architecture comprised nine layers, including three Conv2D layers with ReLU activation, MaxPooling2D, a flatten layer, fully connected dense layers, as well as a softmax output for classification of multi-class. MFCC feature representations were used to train all models through supervised learning with labelled data. All models were trained on MFCC feature representations using supervised learning with labelled data. The study shows deep learning models can effectively classify Assamese traditional songs, with CNN achieving state-of-the-art performance for Borgeet. Furthermore, it provides an entirely new perspective by connecting the Brajavali ASR research with Borgeet's acoustic modelling. Future improvements could involve integrating domain-specific knowledge, advanced feature extraction, and data augmentation (e.g., pitch/tempo variations, noise addition). Furthermore, by adjusting previously trained models to the distinctive features of Assamese compositions such as Borgeet, Bihu, and Kamrupiya Lokogeet, transfer learning might enhance performance more significantly.

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